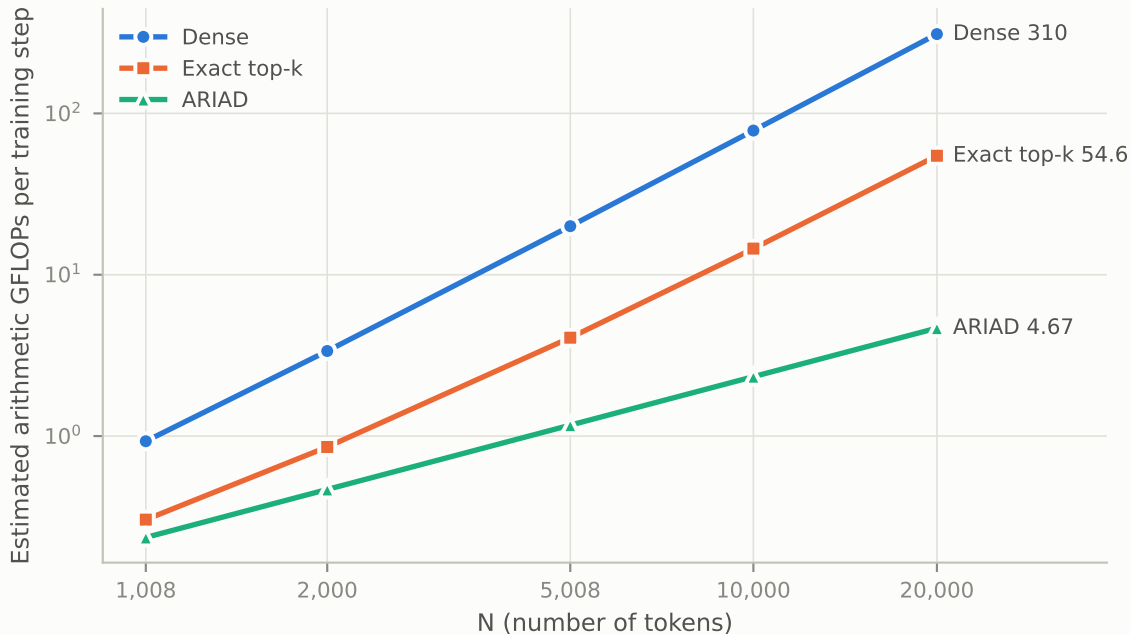


Estimated training FLOPs vs N (dual-space QK, k=32, sparse top-32 teacher)



Log-log slope: Dense 1.95, Exact top-k 1.74, ARIAD 1.00. ARIAD scores C=480 slots/row/step (QQ 176 + KK 176 + QK 128). Cost-model estimate, not measured; excludes softmax, top-k, dedup and memory traffic.